

$$(X^T X)^{-1} = ((X^T X)^{-1})^T \text{ by symmetric prop.}$$

$$S_x = \sqrt{\sum (x_i - \bar{x})^2}$$

$$S_{xx} = \sum (x_i - \bar{x})^2$$

min $RSS(\beta) \rightarrow$ differentiate w.r.t. β , set zero, and solve

$$\hat{COV}(A+B, C) = \hat{COV}(A, C) + \hat{COV}(B, C)$$

$$\hat{COV}(X, Y) = \frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$COV(AY, BY|X) = A COV(Y|X) B^T$$

$$r_{z_1, z_2} = \frac{\sum (z_1 - \bar{z}_1)(z_2 - \bar{z}_2)}{\sqrt{\sum (z_1 - \bar{z}_1)^2} \sqrt{\sum (z_2 - \bar{z}_2)^2}}$$

$$COV(z_1, z_2)$$

$$COV(z_1, z_2) = E\{(z_1 - E(z_1))(z_2 - E(z_2))\} = COV(z_1, z_2); \text{Corr}(z_1, z_2) = \frac{COV(z_1, z_2)}{\sqrt{Var(z_1)} \sqrt{Var(z_2)}}$$

$$COV(z_1, z_2) = 0 \Rightarrow z_1, z_2 \text{ ind.}; E(a z_1 + b) = a E(z_1) + b; Var(a z_1 + b) = a^2 Var(z_1)$$

$$E(z_1 + \dots + z_m) = E(z_1) + \dots + E(z_m); Var(z_1 + \dots + z_m) = Var(z_1) + \dots + Var(z_m) + 2 \sum COV(z_i, z_j)$$

orthogonal: $x \cdot y = 0 \Leftrightarrow x \perp y$

$$(AB)^T = B^T A^T; (A^T)^T = A; (A^{-1})^T = (A^T)^{-1} = A^{-T} = A^{-1}$$

$$COV(Z) = \begin{pmatrix} Var(z_1) & COV(z_1, z_2) \\ COV(z_2, z_1) & Var(z_2) \end{pmatrix}$$

$$Z \in \mathbb{R}^n; B \in \mathbb{R}^{n \times m}$$

$$COV(AZ+B) = A COV(Z) A^T + B B^T$$

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Ch.1 - Introduction

Review

$$X \cdot Y = x_1 y_1 + \dots + x_n y_n$$

Y-dependent (response) var

$$\|X\|_2 = \sqrt{x_1^2 + \dots + x_n^2} = \sqrt{X^T X}$$

X-predictor (ind.) var(s)

$$\|X\|_F = \left(\sum_{i=1}^n |x_i|^2 \right)^{1/2}, p > 0$$

$X_1, \dots, X_p$ $p=1$: SLR $p>1$: MLR

$$E(z_1 + \dots + z_m) = E(z_1) + \dots + E(z_m); Var(z_1 + \dots + z_m) = Var(z_1) + \dots + Var(z_m) + 2 \sum COV(z_i, z_j)$$

1: multi-dim. $\Rightarrow$ multivariate regression

$$COV(Z) = \begin{pmatrix} Var(z_1) & COV(z_1, z_2) \\ COV(z_2, z_1) & Var(z_2) \end{pmatrix}$$

Association $\Rightarrow$ causation

$$COV(AZ+B) = A COV(Z) A^T + B B^T$$

reg. analysis alone cannot establish causation

$$COV(AZ+B) = A COV(Z) A^T + B B^T$$

cover: X-discrete, cont., categorical

$$COV(AZ+B) = A COV(Z) A^T + B B^T$$

Y: continuous (linear)

$$COV(AZ+B) = A COV(Z) A^T + B B^T$$

Binary (logistic)

$$COV(AZ+B) = A COV(Z) A^T + B B^T$$

Discrete (Poisson)

$$COV(AZ+B) = A COV(Z) A^T + B B^T$$

GLM

$$COV(AZ+B) = A COV(Z) A^T + B B^T$$

random component

$$COV(AZ+B) = A COV(Z) A^T + B B^T$$

linear (systematic) comp.

$$COV(AZ+B) = A COV(Z) A^T + B B^T$$

link function

$$COV(AZ+B) = A COV(Z) A^T + B B^T$$

$f(x) = \beta_0 + \beta_1 x$, β non-linear

$$COV(AZ+B) = A COV(Z) A^T + B B^T$$

$f^*(x) = \ln f(x) = \ln(\beta_0 + \beta_1 x)$

$$COV(AZ+B) = A COV(Z) A^T + B B^T$$

linear

$$COV(AZ+B) = A COV(Z) A^T + B B^T$$

$\hat{\beta}$ Properties

$$COV(AZ+B) = A COV(Z) A^T + B B^T$$

treat $\hat{\beta}$ random, assume X: observed fixed, multivariate

$$COV(AZ+B) = A COV(Z) A^T + B B^T$$

$\hat{\beta} = \beta + (X^T X)^{-1} X^T \epsilon$

$$COV(AZ+B) = A COV(Z) A^T + B B^T$$

$E(\hat{\beta}) = \beta$

$$COV(AZ+B) = A COV(Z) A^T + B B^T$$

$COV(\hat{\beta}) = V(\hat{\beta})$

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Ch.2: Estimation

Linear regression

SLR

$$p=1 \rightarrow y_i = \beta_0 + \beta_1 x_i + \epsilon_i, i=1, \dots, n$$

3) ϵ_i 's are ind. (uncorrelated) 4) Normality

$$E(Y|X) = \beta_0 + \beta_1 X$$

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$$E(Y|$$

